Exact Integral-Jet Evaluation of a Rational Noncommutative Hitting Point

Minimal Integral Scaling, Formula-Aware Costs, and Arithmetic Certificates

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Abstract

The source manuscript constructs one rational matrix tuple detecting bounded-size, division-free noncommutative formulas over every characteristic-zero field. Its global guarantee combines faithful iterated integrals with a differential system multiplicity bound; neither ingredient follows from finite moment inversion. We give an exact linear-time action for each integral-jet matrix, then prove a minimal integral diagonal scaling with weights $k!\,\mathrm{lcm}(1,\dots,n)^k$. This yields division-free evaluation of the detecting column of a known formula in $O(sd)$ arithmetic operations, without constructing the dense tuple. The scaling also supplies coordinate-specific denominator bounds, independent of the number of variable occurrences, and certified modular and residual-to-zero tests for rational coefficients. Detailed proofs preserve product order, the constant unit, the source's exceptional one-dimensional branch, and the distinction between a known formula and a matrix black box. Worked cancellation and small-characteristic examples explain both the gains and the limitations. Classical triangular algebra and modular reconstruction are credited; no smaller universal hitting dimension or new general identity test is claimed. Contribution assessment: C3/5, a notable source-specific exact evaluation and arithmetic-certification improvement, without a new multiplicity bound or independent global derandomization result.

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1 Problem, Domains, and What Is Improved

Let $n, s \geq 1$. A formula is a rooted binary tree with variable or scalar leaves and internal $+$ or ordered multiplication gates. Its size counts all gates, including leaves. Scalars lie in a field $\mathbb{F}$ of characteristic zero, and the computed polynomial lies in the free associative algebra $\mathbb{F}\langle x_1, \ldots, x_n\rangle$. Scalar evaluation at matrices means cI. No inverse gates, commutativity relations, or shared subcomputations are assumed. The source [1] claims a single tuple in $\mathrm{Mat}_d(\mathbb{Q})^n$, independent of the formula and its coefficients, that detects every nonzero size-at-most-s formula after scalar extension to any such $\mathbb{F}$. For $ns \geq 2$ its convenient dimension is

$$d = 2ns^2. \tag{1.1}$$

The case $n = s = 1$ uses $d = 1$ and $T_1 = [1]$, not the zero truncation of an integration operator.

This report separates three matters. The source's *hitting theorem* requires a faithful word encoding and a uniform order-of-vanishing theorem. An *evaluation algorithm* requires a representation of the formula or a specified oracle. An *arithmetic certificate* additionally requires exact coefficients or effective height and error bounds. Improvements in the latter two do not reprove or strengthen the first.

The local contribution is a complete structured implementation with a sharp restriction-specific scaling theorem and a denominator ledger that avoids charging the matrix denominator once for every variable occurrence. It is not a claim of historical priority for forward substitution, diagonal conjugation, or the Chinese remainder principle. Applicability L3 is a separate classification, not the contribution score.

2 The Source Chain and Its Independent Obligations

2.1 Ordered integral jets

For a formal series $a \in \mathbb{F}[[z]]$ define

$$J_i a(z) = \int_0^z \frac{a(t)}{t+i}\,dt, \qquad 1 \le i \le n.$$

Integration is termwise with zero constant term; division by positive integers is legitimate in characteristic zero. There is no analytic convergence requirement for this definition. Set $h_\epsilon = 1$ and $h_{iw} = J_i h_w$, where i is the leftmost letter of a word. Over $\mathbb{C}$, these iterated word series converge on $|z| < 1$; an arbitrary formal input need not converge. Thus the rightmost operator acts first.

The order-raising property $J_i(z^d\mathbb{F}[[z]]) \subset z^{d+1}\mathbb{F}[[z]]$ makes the following finite quotient exact:

$$V_d = \mathbb{F}[[z]]/(z^d), \qquad \pi_d a = (a_0, \dots, a_{d-1})^t, \qquad e_0 = (1, 0, \dots, 0)^t.$$

Its matrices, indexed from 0 to $d-1$, are

$$(T_i)_{rq} = \begin{cases} \dfrac{(-1)^{r-q-1}}{r\, i^{r-q}}, & q < r, \\ 0, & q \ge r. \end{cases} \tag{2.1}$$

Expansion of $(z+i)^{-1}$ and one termwise integration prove $\pi_d J_i = T_i \pi_d$. Consequently, for every word polynomial f,

$$f(T_1, \dots, T_n)e_0 = \pi_d h_f, \qquad h_f = \sum_w f_w h_w. \tag{2.2}$$

The empty word contributes $f_\epsilon e_0$. The map $f \mapsto h_f$ is linear, not an algebra homomorphism into the ordinary commutative series algebra. Replacing ordered composition by multiplication of the h_i loses commutators.

2.2 Faithfulness is not a consequence of truncation

The source gives a direct monodromy argument, as well as references to classical hyperlogarithm independence [3]. Its finite-dimensional mechanism is worth recording because it prevents a false dimension argument. Fix a word-length cutoff D and use the truncated free algebra with basis X_w, $|w| \le D$, and augmentation ideal $\mathfrak{m}$. The germ

$$U(z) = \sum_{|w| \le D} h_w(z) X_w, \qquad U' = \left(\sum_i \frac{X_i}{z+i}\right) U, \quad U(0) = 1$$

continues along paths avoiding $-1, \dots, -n$. A loop around $-i$ has endpoint $Q_i = 1 + 2\pi\mathrm{i} X_i \pmod{\mathfrak{m}^2}$. Products $(Q_{i_1} - 1)\cdots(Q_{i_k} - 1)$ have leading degree $(2\pi\mathrm{i})^k X_{i_1}\cdots X_{i_k}$. These elements and 1 form a triangular basis of the finite free algebra. Loop endpoints span it, because products of endpoints are again endpoints with the appropriate concatenation order. A linear relation among the h_w annihilates every endpoint, hence the whole algebra, so all its coefficients vanish.

For any finite word set, the Taylor coefficients are rational. Independence over $\mathbb{C}$ supplies a nonsingular finite Taylor minor over $\mathbb{Q}$: the descending kernels of finite coefficient-prefix matrices stabilize in a finite-dimensional space, with zero intersection. That nonzero rational determinant remains nonzero over every characteristic-zero extension field. No embedding of an arbitrary $\mathbb{F}$ into $\mathbb{C}$ is assumed. This minor argument establishes faithfulness but does not give the uniform prefix length required for the hitting theorem.

2.3 Paths and the differential system

A formula has an acyclic affine-linear path representation with $m \le 2s$ vertices. The source constructs two vertices for a leaf, adds a source and a sink for an addition, and joins product subgraphs in their given order. After a topological ordering, let E be the constant-edge matrix and N_i the x_i-edge matrix. All are strictly upper triangular. Put

$$Q = (I - E)^{-1}, \quad B_i = N_i Q, \quad u = e_{\text{source}}^t Q, \quad v = e_{\text{sink}}.$$

Expanding $Q = I + E + \cdots + E^{m-1}$ enumerates constant-edge stretches and gives $f_w = uB_w v$. Products of m of the B_i vanish. Therefore

$$G = \sum_{|w|<m} B_w h_w, \quad G(0) = I, \quad G' = \left(\sum_i \frac{B_i}{z+i} \right) G, \quad h_f = uGv.$$

Writing $q(z) = \prod_i (z+i)$ expresses this as $G' = PG/q$, with $q(0) = n! \neq 0$, $\deg q = n$, and $\deg P \le n-1$.

2.4 The precise multiplicity bound

The source provides a self-contained formal proof of the needed Moura-type estimate [2]. We summarize its algebraic chain rather than count it as a new Ram theorem. Let $G \in \mathrm{GL}_m(\mathbb{F}[[z]])$, $G' = PG/q$, with the preceding degree and regularity bounds, and constant row and column u, v. For nonzero $y = uGv$,

$$\operatorname{ord}_0 y \le (n-1)\binom{m}{2} + m - 1. \tag{2.3}$$

Indeed, put $h = uG$ and let its component span have dimension k. Rows R_j satisfying $h^{(j)} = R_j G$ obey

$$R_j = Z_j / q^j, \qquad Z_{j+1} = qZ_j' - jq'Z_j + Z_j P, \qquad \deg Z_j \le j(n-1).$$

For the k rows $R_0, \ldots, R_{k-1}$, a nonzero maximal minor has denominator $q^{\binom{k}{2}}$ and numerator degree at most $(n-1)\binom{k}{2}$. The minimum valuation of these minors is unchanged by right multiplication by G: Cauchy–Binet gives one inequality over $\mathbb{F}[[z]]$, and G^{-1} gives the reverse.

A basis of the component span with strictly increasing orders $a_1 < \cdots < a_k$ exists by coefficient elimination. Its Wronskian has order $\sum_j a_j - \binom{k}{2}$, since the leading falling-factorial determinant is a nonzero Vandermonde determinant in characteristic zero. The minimum minor valuation is this order. Since $a_j \ge j - 1$, every nonzero element of the span has order at most $a_k \le (n-1)\binom{k}{2} + k - 1$. Increasing k to m proves (2.3).

Combining faithfulness, the path representation, and (2.3) gives the source's tighter path-program cutoff

$$d_m = (n-1)\binom{m}{2} + m. \tag{2.4}$$

It is already in the source. For formulas, $m \le 2s$ implies $d_m \le 2ns^2$, since the difference at $m = 2s$ is $s(2s + n - 3) \ge 0$. Neither (2.4) nor this simplification is a local improvement. The full source modules were literally read; that is not a formal verification of their entire dependency literature, nor of related rational-formula or positive-characteristic companion manuscripts.

3 Linear-Time Action and Residual Certification

Proposition 3.1 (Bidiagonal realization)**.** *For $a = (a_0, \dots, a_{d-1})^t$, $y = T_i a$ is characterized by*

$$y_0 = 0, \qquad iky_k + (k-1)y_{k-1} = a_{k-1} \quad (1 \le k < d). \tag{3.1}$$

It uses $O(d)$ field operations and $O(d)$ storage including the output, with no dense matrix. Over $\mathbb{R}$ or $\mathbb{C}$,

$$\|T_i\|_{\infty\to\infty} = \max_{1\le k<d} \frac{1}{k}\sum_{\ell=1}^{k} i^{-\ell} \le \frac{1}{i} \le 1 \tag{3.2}$$

when $d \ge 2$; for $d = 1$ the truncated integration matrix is zero.

Proof. The equation $(z+i)y' = a$ and $y(0) = 0$ give (3.1) coefficient by coefficient. Alternatively, adjacent rows of (2.1) cancel all but a_{k-1}. The nonzero coefficient ik gives uniqueness and forward substitution. The induced infinity norm is the maximum absolute row sum, giving (3.2). □

Writing S for multiplication by z on V_d and $D^+ = \mathrm{diag}(0, 1, 1/2, \dots, 1/(d-1))$ also gives

$$T_i = D^+ S(iI + S)^{-1}, \qquad (iI + S)^{-1} = \sum_{j=0}^{d-1} (-1)^j i^{-j-1} S^j.$$

This is a finite nilpotent inverse, with no infinite-series convergence hypothesis. D^+ is the inverse Euler operator on the zero-constant subspace, not an inverse derivative on all formal series.

Proposition 3.2 (Exact residual transfer)**.** *Let $d \ge 2$. Suppose a computed vector $\widehat{y}$ has $\widehat{y}_0 = 0$ and define $r_k = ik\widehat{y}_k + (k-1)\widehat{y}_{k-1} - a_{k-1}$. Then, for $e = \widehat{y} - T_i a$,*

$$e_k = \frac{1}{k}\sum_{j=1}^{k} \frac{(-1)^{k-j} r_j}{i^{k-j+1}}, \qquad |e_k| \le \frac{1}{k}\sum_{j=1}^{k} \frac{|r_j|}{i^{k-j+1}}. \tag{3.3}$$

In particular $\|e\|_\infty \le \max_j |r_j|/i$.

Proof. Subtract (3.1). The error solves the same recurrence with input $(r_1, \dots, r_{d-1}, 0)$, so (2.1) gives the identity and the row-sum bound. If the computed constant term is not zero, it must be separately corrected or enclosed; the differential equation alone does not fix it. □

This is an a posteriori bound for certified residuals, not a claim that floating-point residuals are exact. Rounding in their computation and uncertainty in a require enclosures. A small output or residual is not an identity certificate over fields with unrestricted coefficients.

4 A Minimal Integral Gauge

For the rest of the operator construction let $d \ge 2$, and set

$$L = \mathrm{lcm}(1, \dots, n), \quad \gamma_k = k!L^k, \quad \Gamma = \mathrm{diag}(\gamma_0, \dots, \gamma_{d-1}), \quad U_i = \Gamma T_i \Gamma^{-1}. \tag{4.1}$$

All are rational matrices and $\gamma_0 = 1$.

Theorem 4.1 (Integral recurrence and restricted minimality)**.** *Every U_i has integer entries. For an input b, its output $c = U_i b$ is given over any commutative coefficient ring by*

$$c_0 = 0, \qquad c_k = \frac{L}{i}(b_{k-1} - (k-1)c_{k-1}) \quad (1 \le k < d), \tag{4.2}$$

where L/i is a precomputed integer, not a ring division. The entries are

$$(U_i)_{rq} = (-1)^{r-q-1}\frac{(r-1)!}{q!}\left(\frac{L}{i}\right)^{r-q} \quad (q < r). \tag{4.3}$$

If $\Lambda = \mathrm{diag}(\lambda_0, \ldots, \lambda_{d-1})$ has positive rational entries, $\lambda_0 = 1$, and every $\Lambda T_i \Lambda^{-1}$ is integral, then λ_k/γ_k is a positive integer for every k. Thus (4.1) is coordinatewise minimal by divisibility among normalized positive rational diagonal gauges making the whole tuple integral.

Proof. Substitute $a = \Gamma^{-1}b$, $y = \Gamma^{-1}c$ in (3.1), and use $\gamma_k/\gamma_{k-1} = kL$. This proves (4.2), hence integrality. Direct conjugation gives (4.3). For minimality, the $(k, k-1)$ entry of the conjugated T_i is $\lambda_k/(ki\lambda_{k-1})$. Its integrality for every i says that $t_k = \lambda_k/(k\lambda_{k-1})$ is an integer divisible by $1, \ldots, n$, since the $i = 1$ condition first makes t_k integral. Therefore $\lambda_k = kLa_k\lambda_{k-1}$ for a positive integer a_k. Induction from $\lambda_0 = 1$ yields $\lambda_k = \gamma_k \prod_{j=1}^{k} a_j$, as required. □

The minimality is not an optimality theorem over arbitrary changes of basis, arbitrary integer hitting tuples, or bit-efficient encodings. Negative or complex gauges and non-diagonal transformations are outside its stated class. The result explains a real arithmetic cost: denominator removal in this diagonal class necessarily brings factorial and least-common-multiple growth.

Corollary 4.2 (The integral tuple inherits exactly the old guarantee)**.** *For every characteristic-zero field and every word polynomial,*

$$f(U) = \Gamma f(T)\Gamma^{-1}, \qquad f(U)e_0 = \Gamma f(T)e_0.$$

For the source's bounded formula class and cutoff, the latter vector is zero if and only if $f = 0$. The special branch $n = s = d = 1$ retains $U_1 = [1]$.

Proof. Conjugation respects addition, ordered multiplication, and scalar units. Also $\Gamma e_0 = e_0$ and Γ is invertible in every characteristic-zero field. Invoke the source chain of Section 2, not integrality by itself. □

4.1 A sharp coordinate denominator, not a per-product penalty

Theorem 4.3 (Universal coordinate lattice)**.** *For every $f \in \mathbb{Z}\langle x_1, \ldots, x_n\rangle$ and $0 \le k < d$,*

$$[f(T)e_0]_k \in \gamma_k^{-1}\mathbb{Z}. \tag{4.4}$$

Among positive integers, γ_k is the least common denominator that works for all such integer word polynomials at coordinate k. The sharpness statement concerns unrestricted integer word polynomials, not just formulas of size at most a fixed s.

Proof. The integer tuple gives $f(U)e_0 \in \mathbb{Z}^d$, and $f(U)e_0 = \Gamma f(T)e_0$, proving the inclusion. For $k = 0$, the constant polynomial 1 proves minimality. For $k \ge 1$, take $f = x_i^k$. Since each application of J_i raises order, its leading coefficient is the product $1/i, 1/(2i), \ldots, 1/(ki)$, namely $1/(k!i^k)$. A universal integer denominator must therefore be divisible by every $k!i^k$, whose least common multiple is $k!L^k = \gamma_k$. □

This is stronger than multiplying a denominator bound for the entire matrix once for each variable leaf. It makes the coordinate index, rather than the number of factors in a syntactic product, control the variable-induced denominator. Its proof uses the source's particular integration coefficients; a generic invertible moment matrix does not supply this lattice. Sharpness is not an efficient construction of a bounded-formula worst case: the witness x_i^k has size $2k-1$.

5 Formula-Aware Evaluation and Its Cost Model

Theorem 5.1 (Ordered vector interpreter)**.** *Given a division-free formula F of size s and a vector $b \in \mathbb{F}^d$, one can compute $F(T)b$ in $O(sd)$ field operations without materializing any d-by-d matrix. The same assertion holds for U, using (4.2). A depth-first implementation uses $O(dh+s)$ stored field elements and node data, where $h \le s$ is the tree height. In particular, the detecting column is obtained by setting $b = e_0$.*

Proof. At a scalar leaf return cb; at a variable leaf use the corresponding linear-time recurrence. At addition return

$$\mathrm{Apply}(F_1, b) + \mathrm{Apply}(F_2, b).$$

At an ordered product return

$$\mathrm{Apply}(F_1, \mathrm{Apply}(F_2, b)). \tag{5.1}$$

Induction proves equality with the matrix action. Each node of the formula tree is visited once in a call: the two child calls are on disjoint subtrees, including at multiplication. Each node costs $O(d)$ operations in addition to its children, proving the operation count. A depth-first traversal stores an input or saved partial output at each active level, giving the stated memory bound. □

The proof deliberately says *tree*. For a shared circuit, a gate may be called on different vectors by different parents. Memoizing a single vector per gate is unsound, and expanding the circuit into a tree can be exponential. Nor does this argument handle inverse gates: the matrices T_i and U_i are singular.

Requested object	Permitted local procedure	Scope of the cost
One variable action	Bidiagonal recurrence	$O(d)$ arithmetic operations
Known formula, one vector	Ordered tree interpreter	$O(sd)$ operations; not an oracle simulation
Known formula, full matrix	Apply to all d basis vectors	$O(sd^2)$ operations and d^2 output positions
Explicit rational tuple	Source entry construction	$O(nd^2)$ integer operations on bounded-bit integers
Arbitrary matrix black box	Submit a full admissible tuple	Oracle interface and output cost remain separate

The dense source tuple has $\Theta(nd^2)$ nonzero entries for $d \ge 2$. A recurrence description is a succinct operator representation, not an explicit-output construction of those entries in $O(nd)$ time. A black box that returns the full matrix has its own representation and output costs; the source's one-query statement does not assert that the query itself costs $O(sd)$. A specifically supplied vector-action oracle would be a different interface.

5.1 Entry growth of the integral alternative

Let

$$B_\Gamma = 1 + \lceil \log_2 \gamma_{d-1} \rceil .$$

Every entry of U_i has absolute value at most γ_{d-1}. For example, (4.3) is bounded by γ_r because $(r-1)!/q! \le r!$ and $L^{r-q} \le L^r$. Thus

$$B_\Gamma = O\big(d \log(d+1) + d \log(L+1)\big), \qquad L \le n!, \tag{5.2}$$

an elementary polynomial bit bound. The source's original rational entry bound is only $O(\log(d+1) + d\log(n+1))$. Removing all generator divisions can therefore increase entry size. There is no blanket bit-speedup claim. The constants L/i may be computed once by exact integer division and then reduced modulo the desired moduli. Recurrence evaluation itself requires no division, including modulo primes dividing L.

6 Rational Coefficients and Exact Decision Budgets

Now assume every scalar leaf is presented as p/q with $p \in \mathbb{Z}$, $q \in \mathbb{Z}_{>0}$. Reduction to lowest terms is optional. Let t be the number of scalar leaves, a the number of addition gates, and assume $|p|, q \le 2^H$ for all scalar leaves. The arbitrary-field hitting theorem does not need this hypothesis; the following effective certificates do.

6.1 Denominator clearing without division in the coefficient ring

Associate to every subtree g a positive integer Q_g and an integer noncommutative polynomial $N_g = Q_g f_g$ as follows:

$$\begin{aligned} &(Q_x, N_x) = (1, x), \qquad (Q_{p/q}, N_{p/q}) = (q, p), \\ &Q_{g_1+g_2} = Q_{g_1} Q_{g_2}, \quad N_{g_1+g_2} = Q_{g_2} N_{g_1} + Q_{g_1} N_{g_2}, \\ &Q_{g_1 g_2} = Q_{g_1} Q_{g_2}, \quad N_{g_1 g_2} = N_{g_1} N_{g_2}. \end{aligned} \tag{6.1}$$

These are identities over $\mathbb{Q}\langle x \rangle$; the resulting numerator program uses only integer constants, addition, and ordered multiplication. Let $Q = Q_F = \prod_{\text{scalar leaves}} q$. To apply $N_F(U)$, the addition case of the vector interpreter uses $Q_{g_2} \operatorname{Apply}(N_{g_1}, b) + Q_{g_1} \operatorname{Apply}(N_{g_2}, b)$. Every original node still causes only a bounded number of vector operations. The Q_g can be precomputed once in $O(s)$ integer operations.

Define a computable nonnegative majorant by

$$\begin{aligned} W_x = 1, \quad W_{p/q} = |p|, \\ W_{g_1+g_2} = Q_{g_2} W_{g_1} + Q_{g_1} W_{g_2}, \qquad W_{g_1 g_2} = W_{g_1} W_{g_2}. \end{aligned} \tag{6.2}$$

This is a coefficient-ℓ^1 majorant, not an assumed absence of cancellation.

Theorem 6.1 (Coordinate heights and certified modular decisions)**.** *For $z = N_F(U) e_0 = Q \Gamma f_F(T) e_0$ one has, for every $0 \le k < d$,*

$$z_k \in \mathbb{Z}, \qquad |z_k| \le \gamma_k W_F, \qquad W_F \le 2^a \prod_{\text{scalar leaves}} \max(|p|, q) \le 2^{a+Ht}. \tag{6.3}$$

Let $m_1, \dots, m_\ell \geq 2$ be pairwise coprime integers and $P = \prod_j m_j$. Any nonzero coordinate of $z \bmod m_j$ proves $f_F \neq 0$ over $\mathbb{Q}$. If all coordinates vanish modulo every m_j and

$$P > \gamma_{d-1} W_F, \tag{6.4}$$

then $f_F = 0$, provided the source size promise and its sufficient cutoff hold. Coordinate-dependent products $P_k > \gamma_k W_F$ also suffice if each coordinate has been checked with its own pairwise-coprime list.

Proof. Induction in (6.1) proves $N_g = Q_g f_g$. The coefficient-ℓ^1 norm of N_g is bounded by W_g, using the triangle inequality and submultiplicativity of that norm in a free algebra. Since each T_i is a contraction in the induced infinity norm, every word matrix is a contraction. Hence $\|N_F(T)e_0\|_\infty \leq W_F$. Conjugation and integrality give the first two assertions of (6.3).

For the last assertion let P_g be the product of $\max(|p|, q)$ over scalar leaves of g. Then $Q_g \leq P_g$. Multiplication preserves $W_g \leq 2^{a_g} P_g$. At addition, the two terms are at most $P_{g_1} P_{g_2} (2^{a_{g_1}} + 2^{a_{g_2}})$, which is bounded by $2^{a_{g_1} + a_{g_2} + 1} P_{g_1} P_{g_2}$. This proves the height estimate.

All arithmetic in $N_F(U)$ can be performed in $\mathbb{Z}/m_j\mathbb{Z}$ because the recurrence and numerator program are division-free. A nonzero residue cannot come from zero z. If all residues vanish, each z_k is divisible by P. Under (6.4), a nonzero multiple of P contradicts $|z_k| \leq \gamma_k W_F \leq \gamma_{d-1} W_F$. Thus $z = 0$, and the source's detecting-column theorem gives $f_F = 0$. The coordinate-specific version uses the same argument separately. □

The zero test needs $P > M$, not $P > 2M$, for an absolute height bound M. The latter is the familiar sufficient threshold for unique signed reconstruction. This elementary distinction is already explained in the private comparison [8]; it is not claimed as a new theorem here. The new ingredient in its present application is the proved coordinate height $\gamma_k W_F$ rather than an unspecified coefficient bound.

6.2 Costs, small primes, and extension fields

The numerator interpreter costs $O(sd)$ ring operations per modulus after the integer constants have been reduced. For b-bit moduli, schoolbook arithmetic gives $O(sd\,b^2)$ bit operations for that evaluation, excluding constant reduction, modulus generation, and the precomputation of L and Q_g. These are separate finite tasks. Direct exact integer evaluation also has polynomial bit cost: a conservative intermediate bit bound is

$$O\big(s(B_\Gamma + H + \log(d+1))\big).$$

To see this without relying on cancellation, bound each generator matrix norm by $d\gamma_{d-1}$, each scalar numerator by 2^H, and each denominator weight by the product of its subtree denominators. Along a tree evaluation, the contributing subtrees are disjoint; addition costs at most another factor 2 per gate. The recurrence's internal multipliers have at most $O(\log(d+1) + \log(L+1))$ bits and fit this conservative bound. The much smaller bound (6.3) concerns the final detecting column.

The ring procedure permits moduli dividing L, Q, or k! because it reduces the cleared integer numerator, not the original rational formula with illegal inversions. It does *not* assert that $U \bmod p$ is a universal hitting tuple in characteristic p. For $n = 2$, $L = 2$, every nonzero entry of U_1 is divisible by 2. Thus the nonzero polynomial x_1 evaluates to zero modulo 2. An isolated zero residue says nothing beyond divisibility. The product-of-moduli height condition recovers an exact statement over $\mathbb{Q}$, not a new theorem over finite fields.

Over an arbitrary characteristic-zero extension, the vector interpreter and conjugation identities remain valid as field-operation statements. The rational height, integer lattice, and modular zero conclusions do not extend to arbitrary coefficients without an effective coefficient ring and its own separation theory. In particular, there is no universal positive numerical zero threshold for $f = \varepsilon x_1$ with unrestricted nonzero ε.

6.3 A posteriori enclosures for a whole known formula

For completeness, certified local errors can be propagated without using a dense matrix. Let M_g be an upper bound for $\|f_g(T)\|_{\infty\to\infty}$ obtained by $M_x = 1/i$, $M_c = |c|$, addition of bounds at $+$, and multiplication of bounds at a product. An inexact vector interpreter may produce different local error enclosures on different inputs. Write $E_g(b)$ for a certified output-error bound relative to the exact input b. At addition, sum the two child errors and the certified rounding error of vector addition. At a product, compute an approximate right output $\widehat{v}$ and use

$$E_{g_1 g_2}(b) \le E_{g_1}(\widehat{v}) + M_{g_1} E_{g_2}(b). \tag{6.5}$$

The triangle inequality proves this by inserting $f_{g_1}(T)\widehat{v}$. Variable-leaf errors follow from Proposition 3.2; scalar errors require the stated coefficient and arithmetic enclosures. This recursion preserves the actual right-first order.

Corollary 6.2 (An exact-zero enclosure gate)**.** *For rational F as above, suppose $\widehat{y}_k$ encloses the true $y_k = [f_F(T)e_0]_k$ with certified errors $\epsilon_k \ge 0$. If for every $k < d$*

$$|\widehat{y}_k| + \epsilon_k < \frac{1}{Q\gamma_k}, \tag{6.6}$$

then $y = 0$, and hence $f_F = 0$ under the source size and cutoff promise. A certified inequality $|\widehat{y}_k| > \epsilon_k$ at any coordinate proves nonidentity without that cutoff promise.

Proof. Theorem 4.3 applied to N_F places y_k in $(Q\gamma_k)^{-1}\mathbb{Z}$. A nonzero point in that lattice has absolute value at least the right side of (6.6). For nonidentity, the enclosure simply excludes zero in one exactly defined evaluation. □

No unknown inverse norm or empirical numerical rank appears here. However, γ_k can be large, so the required absolute precision can be severe. Transforming to the integral gauge moves that cost into the size of the coordinates; it does not remove it.

7 Worked Certificates and Necessary Counterexamples

7.1 A commutator invisible to a too-short prefix

For $n = 2$, $d = 4$, $L = 2$, and $\Gamma = \mathrm{diag}(1, 2, 8, 48)$, the integral matrices are

$$U_1 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ -4 & 2 & 0 & 0 \\ 16 & -8 & 2 & 0 \end{pmatrix}, \qquad U_2 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 2 & -2 & 1 & 0 \end{pmatrix}.$$

For $C = x_1x_2 - x_2x_1$, direct ordered evaluation gives

$$C(U)e_0 = (0, 0, 0, -2)^t, \qquad C(T)e_0 = (0, 0, 0, -1/24)^t.$$

The first three Taylor coefficients vanish, although C has word degree two and is nonzero. This fixture, also consistent with the source's commutator computation, calibrates the distinction between degree and vanishing order. It is not a claim that $d = 4$ hits the entire formula class. It also detects reversal of the product order and replacement by ordinary commutative multiplication: those changes negate or annihilate the commutator.

7.2 A rational numerator and a bad modular zero

Take

$$f = \frac{1}{6}(x_1x_2 - x_2x_1) + \frac{2}{5},$$

with subtraction implemented by a scalar -1 leaf. The tree denominator is $Q = 30$. The majorant recursion yields $W_F = 22$, and

$$f(T)e_0 = (2/5, 0, 0, -1/144)^t, \qquad Q\Gamma f(T)e_0 = (12, 0, 0, -10)^t.$$

The final-coordinate bound is $48 \cdot 22 = 1056$. Reduction modulo 2 gives the zero column, while modulo 3 the last coordinate is 2. Thus a small prime can miss this particular nonzero numerator even though no illegal modular division occurs. For a zero claim, a product such as $11 \cdot 13 \cdot 17 = 2431 > 1056$ would be a sufficient modulus budget *if all the required residues were zero*. They are not zero for this example. A budget is a condition on a certificate, not a prediction of its outcome.

7.3 What the fixed tuple cannot detect without a size promise

In the integration branch, every T_i is strictly lower triangular and $T_i^d = 0$. Therefore the nonzero word x_i^d is annihilated. Its formula needs $2d - 1$ gates, outside the intended size regime when $d = 2ns^2$. The example explains why faithfulness before truncation and the multiplicity cutoff are indispensable. Small shared circuits can produce high powers by repeated squaring, so replacing formula size by circuit size would invalidate the source chain.

Constant handling supplies another essential test. At $d = 1$, the formal integration quotient gives $T_i = 0$; the source instead chooses [1] in its $n = s = 1$ branch. Only variable and scalar leaves occur in that size-one class, so the choice detects every nonzero leaf. It need not distinguish $x_1 - 1$, which is not a size-one formula.

8 The Ram Connection: Exact Finite Algebra, Not a New Encoding

The fresh original private proof used here is the finite polynomial moment/inversion argument in [9], together with its implementation supplement [10]. Its precise content is as follows. If h is an integrable signed or complex kernel with $\int(1 + |t|^N)|h(t)|\,dt < \infty$, then on polynomials of degree at most N,

$$(Sf)(x) = \int f(x - t)h(t)\,dt = \sum_{j=0}^{N} \frac{(-1)^j\mu_j}{j!} f^{(j)}(x).$$

Taylor's formula is finite, so the absolute moment assumption justifies every integral. Writing $S = \mu_0 I + R$, the nonconstant part strictly lowers degree and $R^{N+1} = 0$. Consequently, if $\mu_0 \neq 0$,

$$S^{-1} = \mu_0^{-1} \sum_{j=0}^{N} (-R/\mu_0)^j.$$

If $\mu_0 = 0$, constants lie in the kernel. These statements are finite algebraic identities; they do not imply a bounded inverse for infinite analytic or physical-signal spaces.

The application to M232 is specific but limited. The nilpotent shift S used in Section 3 raises coefficient order rather than lowering polynomial degree, yet its finite reciprocal algebra is the same elementary mechanism. This supplies an exact implementation vocabulary, not the independence of the word integrals or the source's vanishing-order bound. Moreover, the original implementation supplement distinguishes derivative jets from Taylor coefficients by factorials. In the present report π_d always stores ordinary Taylor coefficients, while $\Gamma\pi_d$ stores L-scaled derivative jets. Losing that factorial would destroy (4.2).

The selected proof passages of [8] explain a second useful restriction: an invertible coordinate change must carry the operations and the unit with it. On a finite polynomial quotient an invertible linear map R has transported product

$$a \star_R b = R((R^{-1}a)(R^{-1}b))$$

and unit $R1$. This is not ordinary coordinatewise multiplication. For M232 the relevant algebra is instead the algebra of endomorphisms: the transformation is $A \mapsto \Gamma A \Gamma^{-1}$, which preserves ordered composition and scalar units, while $\Gamma e_0 = e_0$. The distinction matters because $f \mapsto h_f$ is not multiplicative for ordinary series multiplication.

There is also a difference between a well-scaled physical norm and an integral encoding. The physical coefficient norm of $b = \Gamma a$ is $\|\Gamma^{-1}b\|_\infty$, not $\|b\|_\infty$. In this weighted norm the U_i retain the contractions of the T_i, although their unweighted entries can be large. The diagonal condition number is γ_{d-1} when $d \geq 2$. No floating-point conditioning benefit follows merely from eliminating denominators.

Finally, [8] supplies general lattice and bounded-height modular principles. Here they become implementable through Theorems 4.1, 4.3, and 6.1, which derive the actual source-specific scaling and heights. They are not substitutes for faithful noncommutative encoding. No unread private span or Gram theorem is imported into this proof.

9 Public Literature and Comparative Scope

The October 8, 2026 refresh used only public titles and public URLs. The following distinctions rely on the primary author or publisher records, not on third-party status labels.

Raz–Shpilka [4] give deterministic polynomial-time identity testing when noncommutative formulas or appropriate branching programs are supplied. Thus the existence of a polynomial-time *white-box* test is not new here. Forbes–Shpilka [5] obtain quasipolynomial black-box hitting sets for specified branching-program models, including a reduction covering noncommutative ABPs. Those prior model distinctions remain important.

Lakhani–Mukhopadhyay's July 20, 2026 revision [6] gives integer matrix hitting points parameterized by sparsity, with entry bits $O(\sigma \log n)$ for sparsity σ. Formula size and sparsity are different: a product of k sums $(x_1 + x_2)$ has 2^k words but a linear-size formula. Our integral conjugation is therefore not a claim to have first constructed an integer hitting point. Its specific contribution is a minimal diagonal gauge and an evaluation/certification procedure for the source's formula-size-controlled tuple, with the stated larger entry cost.

Arvind–Chatterjee–Mukhopadhyay [7] treat noncommutative *rational* formulas in a free skew field and give a deterministic quasipolynomial hitting-set result. Inverse gates require defined, invertible values. The singular nilpotent tuple in this report cannot simply be substituted into those

inverse gates. We do not import the separate OpenAI companion claims about rational formulas or positive characteristic.

The hyperlogarithm and multiplicity references [3, 2] identify classical antecedents of the source route. Their current primary abstracts and metadata were refreshed, but their complete external proofs were not independently audited here. The full formal special-case multiplicity argument and direct monodromy proof in the cached M232 source were read. The distinction avoids claiming either that these inputs are newly proved by Ram coordinates or that finite tests certify the global hitting theorem.

10 Read Scope, Tests, and Source Details Retained

All nine cached public text files were read literally in bounded displays: 1,075 lines in total. The five scientific modules comprise 823 of those lines. The remaining files are the main wrapper, proof-route figure source, bibliography, and README. Pre-read and post-read hashes, actual intervals, unread complements, and the complete heading map are recorded in the owned read-scope ledger. The source dependency map is:

File	Literal lines	Mathematical role
`introduction`	1–193	Model, quantifiers, bit construction, prior-work scope
`iterated-integrals`	1–176	Word order, monodromy, rational minor descent to all characteristic-zero fields
`formula-systems`	1–138	Acyclic path representation, constant-edge elimination, differential system
`multiplicity`	1–157	Wronskian valuation and degree bound
`construction`	1–159	Exact tuple, detecting column, bit cost, tighter ABP cutoff

Fresh private reads consist of 235 selected lines of TR-item-317-astra: 1–174, 340–382, and 417–434. Its lines 175–339 and 383–416 were not read for this report. The original Volume 1 chapter was read at lines 1–91 of 510, and its implementation supplement at lines 1–50 of 129. Their remaining 419 and 79 lines, and the uninspected include files, are not claimed as read. The original proof scope is therefore 141 lines, in addition to the selected comparison passages; this is not a whole-corpus audit. Confidential material was not sent to public search services.

No substantive correction to the inspected source proof chains was established. The following valid source details are deliberately retained: the zero integration constant, right-first product order, scalar cI, the special $n = s = 1$ choice, independence over arbitrary characteristic-zero extensions by a rational minor rather than an assumed complex embedding, the sharper existing path-program cutoff, and explicit-output bit cost. The new recurrence and minimal integral/denominator certificates are local deductions from those details, not corrections of a stated source formula.

The independently executable Python file `M232-exact-checks.py` uses exact fractions and unbounded integers. It passes 3,644 finite cases containing 74,524 explicit assertions. These units are not interchangeable: a matrix case contains many entry assertions, including zeros. The group ledger reports both. It checks rational and integral recurrence actions against dense matrices, conjugated entries and row norms, exact residual identities, diagonal-gauge fixtures, sharp coordinate denominators, 144 tree formulas against both dense evaluation and independently expanded word coefficients, 720 modular numerator cases including bad primes, and the displayed counterexamples. Some formula fixtures intentionally use small arbitrary cutoffs and thus check evaluation identities, not universal detection. No parent or independent-review test is included in these author counts. No

performance benchmark, external formal-proof build, or randomized evidence of a global theorem is claimed.

11 Advantages, Costs, and Further Work

The principal advantage is a reproducible bridge from the exact source tuple to an inexpensive *known-formula* column computation. The recurrence eliminates dense generator storage, the integer gauge removes runtime divisions, and the sharp universal coordinate lattice replaces a potentially much worse product-of-denominators ledger. Residual certification can stay in the contractive Taylor coordinates, while modular arithmetic can use the integral coordinates. Both are tied by an exact conjugation identity.

The costs are equally explicit. The lcm and factorial can enlarge integers; the minimality theorem says that this is unavoidable inside the specified positive diagonal class, not universally unavoidable. The full black-box tuple and full output remain large. Coefficient heights and formula-tree access are additional data needed by the arithmetic algorithms. The multiplicity dimension remains unchanged, and short numerical prefixes cannot replace it. The construction is neither a commutative circuit PIT result nor a method for arbitrary rational formulas with inverse gates.

Three concrete research directions remain. First, a non-diagonal integral lattice might lower bit cost while preserving an equally sparse action; the diagonal lower bound does not decide that question. Second, one can investigate denominator bounds restricted simultaneously by coordinate index and formula size. The unrestricted sharpness examples in Theorem 4.3 are not always size-admissible. Third, a vector-action oracle interface or additional structured circuit representation could make the local interpreter usable beyond explicit trees; its semantics, reuse costs, and zero certificate would have to be proved separately. Improving the universal dimension still requires an improvement to the faithful encoding or multiplicity analysis, not simply faster application of the existing matrices.

12 Conclusion

The source's single rational hitting point admits a precise, division-free integral implementation and a sharp coordinate denominator theory. Together they give source-specific vector evaluation, height, modular, and enclosure certificates with proved cost and scope boundaries. The C3 assessment rests on this combined local improvement, especially the minimal integral gauge and the denominator bound independent of variable occurrence count. It does not assign new credit to classical forward substitution or CRT, and it does not assert a new global hitting theorem.

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[8] M. Subbarao, *Ram Calculus and Deterministic Polynomial Identity Testing: Exact Jet Algebras, Structured Span Certificates, and Quantitative Separation*, confidential local TR-item-317-astra, 2026. Only the proof passages specified in Section 10 were used.

[9] M. Subbarao, *Ram Calculus, Volume 1*, revised Chapter 2, *The Spatial-Domain S-Transform*, confidential original local source `ch02-s-transform-rev2.tex`, lines 1–91.

[10] M. Subbarao, *Implementing and checking the finite algebra*, confidential original implementation supplement `ch02-A-implementation.tex`, lines 1–50.